

OmniMed-FL: A Robust Multimodal Federated Learning Framework for Clinical Diagnosis

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Abstract—Clinical diagnosis requires medical images and patient records together, yet most machine learning systems model one modality, and privacy rules such as HIPAA and GDPR block hospitals from pooling records centrally. We propose OmniMed-FL, a multi-modal federated learning framework for five-class clinical condition classification (Normal, Pneumonia, COVID-19, Pleural Effusion, Cardiomegaly). It benchmarks five LLM text encoders, five Vision Transformer (ViT) image encoders and eight VLM fusion strategies under FedAvg with non-IID Dirichlet($\alpha=1.0$) partitioning across 5 hospital clients. An anti-collapse stack of balanced batch sampling, entropy diversity regularization and adaptive early abort prevents majority-class collapse: dips are transient and all 18 variants recover full five-class diversity by epoch 5. DistilBERT reaches F1 0.934 on text, ViT-Base 0.664 on images, and Concat VLM fusion 0.956—a +2.2 point gain over the best unimodal baseline. Federated training retains 99.1% of centralized Macro F1 averaged over the three branches: Fed-VLM (0.956) matches its centralized counterpart and Fed-LLM (0.953) exceeds its own (0.934), while the image-only branch loses 4.8%. Against the 31 published systems compiled in Table IV, Fed-VLM is the strongest federated entry and second overall behind a centralized EfficientNet-B7 (0.980), though those systems use different corpora so the positioning is indicative only. A FAISS-indexed retrieval layer returns the correct condition as its top-1 match for all five probe queries, at cosine similarities of 0.66–0.85.

Index Terms—Multimodal Federated Learning, Vision-Language Models, Clinical Condition Classification, E-Health, Privacy-Preserving AI, Vision Transformer

I. INTRODUCTION

CLINICAL condition classification—distinguishing normal presentations from acute infections, fluid accumulation, and structural cardiac pathology—is central to timely patient intervention. Most automated diagnostic systems rely on a single data type: CNNs or vision transformers over chest radiographs [1], [2], or language models over electronic health records. Neither is sufficient alone. Image-based models overlook clinical context such as elevated BNP levels or a history of viral exposure; text-based models cannot capture radiological findings such as bilateral ground-glass opacities or a pleural fluid meniscus.

Deployment is further constrained because patient data is spread across hospitals, clinics and diagnostic centres, while

HIPAA and GDPR restrict centralised collection of medical records. Federated Learning (FL) offers a practical alternative: institutions train a shared model through locally computed parameter updates without exchanging raw data. Most clinical FL methods, however, remain single-modality and ignore the textual notes clinicians record alongside images [4]–[6]. Physicians draw on both sources, yet existing frameworks either pool sensitive data centrally or model images alone. We therefore developed OmniMed-FL, a federated multimodal framework that lets hospitals train on imaging and textual information together without exposing raw patient data.

A. Contribution

The main *contributions* of this paper are as follows:

- We study how combining medical images with electronic health records improves diagnosis in a federated setting; both modalities together outperform either alone.
- We prepare a controlled clinical dataset in which similar information appears across disease classes, so the model cannot predict from a few obvious keywords and must use both the image and the clinical record.
- We evaluate OmniMed-FL under non-IID distributions reflecting differences between participating hospitals, reaching performance close to centralized training while keeping patient data within each institution.

II. RELATED WORK

A. Medical Image Classification

Early work treated radiology as an image classification problem. Rajpurkar *et al.* [1] and Irvin *et al.* [2] showed the effectiveness of CNN-based chest X-ray classification in centralized settings; later work explored Vision Transformers [7], [8], which capture broader spatial relationships within an image. These approaches classify strongly but rely on centrally collected datasets and ignore the clinical notes that often accompany medical images.

B. Federated Clinical Learning

FL has been studied as a way to train clinical models without moving patient data outside individual institutions. Sheller *et*

TABLE I: Positioning vs. Related Work.

Work	Modality		Framework	
	Img	Text	FL	VLM
CheXNet [1]	✓	×	×	×
Sheller 2020 [4]	✓	×	✓	×
Kaissis 2021 [6]	✓	×	✓	×
BiomedCLIP [9]	✓	✓	×	✓
MedCLIP [11]	✓	✓	×	✓
OmniMed-FL (ours)	✓	✓	✓	✓

al. [4] applied FedAvg in a decentralized medical imaging setting, Dou *et al.* [5] validated a federated model across national sites, and Kaissis *et al.* [6] added differential privacy. Most such approaches are built around a single modality and consider medical images only.

C. Vision-Language Models in Healthcare

Another line of work combines medical images with textual information. BiomedCLIP [9], LLaVA-Med [10] and MedCLIP [11] learn relationships between medical images and their associated reports, using both modalities rather than treating the image independently. These models are generally trained centrally, which makes direct use across hospitals with isolated patient data difficult.

Synthesis: As summarized in Table I, existing work generally addresses either multimodal learning or decentralized training, and comparatively little combines both within one framework. OmniMed-FL addresses this gap by incorporating image–text fusion into a federated learning setup for clinical diagnosis.

III. SYSTEM DESIGN

A. Architecture

Figure 1 shows the overall OmniMed-FL pipeline. Clinical notes are first tokenized using WordPiece with a maximum sequence length of 128 tokens, while chest radiographs are resized to 224×224 and normalized using ImageNet statistics. The text encoder produces a feature vector $\mathbf{h}_t \in \mathbb{R}^{256}$ using the [CLS] representation, whereas the image encoder generates $\mathbf{h}_v \in \mathbb{R}^{512}$ using adaptive average pooling.

The two feature representations are then combined in the VLM module. We evaluate eight fusion methods: Concatenation, Cross-Attention, Gated Fusion, CLIP, Flamingo, BLIP-2, CoCa, and Unified-IO. Each produces a joint representation $\mathbf{h}_f$, passed to a linear classifier for five-class prediction. We included both simple and attention-based methods to test whether the extra computation is justified in a federated hospital setting. As Figure 2 shows, seven of the eight fall within a 1.0-point band (0.946–0.956) and all seven exceed the best unimodal encoder, with plain concatenation highest. Since each variant was run once, that spread is not a defensible ordering: the benefit comes from having both modalities rather than from the fusion operator, which favours the cheapest option. CoCa (0.662) does not converge within this budget.

B. Encoding Specifications

The text branch uses DistilBERT: 6 transformer layers, hidden size 768, 12 attention heads, ≈ 66 M parameters. The first

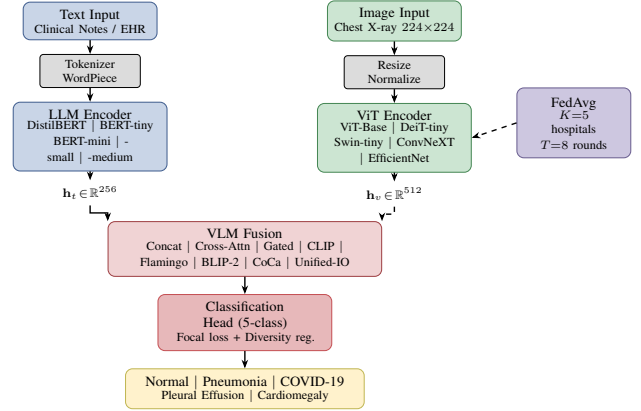


Fig. 1: OmniMed-FL architecture. LLM and ViT encoders process clinical notes and chest X-rays; VLM fusion combines both modalities; FedAvg coordinates $K=5$ hospital clients without transmitting raw patient data.

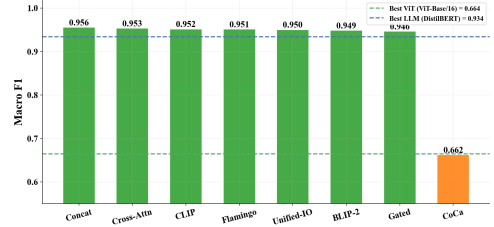


Fig. 2: Benchmark of 8 VLM fusion strategies. Simple Concatenation (Concat) and Gated Fusion provide the best balance of Accuracy and Macro F1.

four layers stay frozen and the rest are fine-tuned on the clinical data, cutting trainable parameters while still adapting to medical terminology and task-specific patterns. The image branch uses ViT-Base/16: 12 layers, hidden size 768, 12 heads, ≈ 86 M parameters. We selected it as the highest-scoring backbone in Figure 3, though its margin over Swin-tiny (0.664 vs. 0.661) is small enough that the two should be treated as equivalent here; the clear separation is between the attention-based backbones and the convolutional ones. Images are split into 16×16 token patches with a learnable [CLS] token prepended. For both modalities a projection head maps the native hidden states to a shared latent space ($d = 256$ or 512) prior to concatenation or cross-attention fusion.

C. Dataset and Clinical Corpus

Image data. We aggregated public chest X-ray collections and mapped them to our five target conditions: the kerememberke pneumonia dataset, Tawsifur’s COVID-19 image set, and the NIH Chest X-ray dataset (filtered from 14 labels to our 5). Where a class lacked enough real samples we supplemented the training set with synthetic, X-ray-style images carrying specific pixel signatures—a bright lower-lobe patch for Pneumonia, a basal opacity gradient for Pleural Effusion. All images are ImageNet-normalized and stored as float16 tensors to halve memory use. The corpus is balanced at 600 samples per class, 3,000 images total, split 80/20 for training and validation.

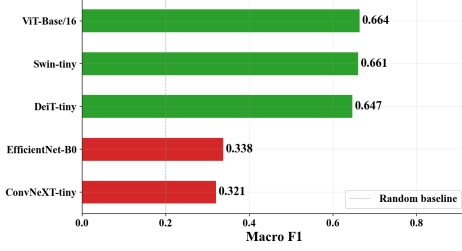


Fig. 3: Comparison of image backbones. The three attention-based backbones cluster at Macro F1 0.647–0.664; the two convolutional baselines fall to 0.32–0.34.

Clinical text data. Because real EHR data is difficult to source publicly, a template engine generates 3,000 synthetic medical reports from vocabulary pools mapped to observations, symptoms and conditions. A 75%/25% mixed-to-clear slot ratio reduces reliance on obvious keywords: mixed samples carry overlapping or cross-class phrasing, making the text less directly indicative of the target class and encouraging use of the full clinical context. All generated text is normalized and simulated patient identifiers removed. Templates keep the pathological description consistent with the findings in the paired X-ray, giving a controlled setting for evaluating how effectively the fusion module combines the two modalities.

What this corpus does and does not support. Because cross-modal alignment is known by construction, every variant sees identical data and differences between variants are attributable to the variant. The corpus is not a substitute for clinical data, and it bounds what the numbers mean. Template notes draw on a smaller, more regular vocabulary than dictated reports and reproduce none of their negation-scope errors, hedging, abbreviation drift or comorbidity structure; the images used to top up under-represented classes are pixel-pattern surrogates, not radiographs. Both make the modalities more separable than they would be in practice, and asymmetrically so—the text branch benefits more—so the modality gap we report is very likely an upper bound on the real one, and the absolute Macro F1 values are ceilings obtained under favourable conditions. What the corpus does support are the *relative* comparisons: across modalities, across fusion strategies, and between federated and centralized training.

D. Experimental Setup

Hyperparameters. Each client trains locally with AdamW, using learning rate 2×10^{-5} for the language encoder and 1×10^{-4} for both the ViT and the fusion head, with linear warmup over the first 10% of training iterations. Each client performs five local epochs ($E = 5$) before sending updated parameters to the server for aggregation. The diversity-regularizer coefficient is $\lambda = 0.1$, set by grid search over 0.01 to 0.5. All experiments use PyTorch 2.4 on NVIDIA A100 GPUs with Automatic Mixed Precision (AMP). Table II gives the complete configuration.

Partitioning. To represent differences in data distributions across hospitals, we divide the dataset among clients with a Dirichlet non-IID partition at concentration $\alpha = 1.0$, which

TABLE II: Global Hyperparameter Configuration.

Parameter	Value
Local Epochs (E)	5
Federated Rounds (T)	8
Batch Size	32
LLM Learning Rate	2×10^{-5}
ViT Learning Rate	1×10^{-4}
Fusion Learning Rate	1×10^{-4}
Weight Decay	0.01
Warmup Steps	100
Diversity Reg. (λ)	0.1
Dirichlet Alpha (α)	1.0
Precision	FP16 (AMP)

TABLE III: Macro F1 Scores Across All 18 Evaluated Variants

Modality	Model Variants (Macro F1)	Peak F1
LLM (text)	BERT-tiny (0.786), BERT-mini (0.873), BERT-small (0.898), BERT-medium (0.911), DistilBERT (0.934)	0.934
ViT (image)	ConvNeXT-tiny (0.321), EfficientNet (0.338), DeiT-tiny (0.647), Swin-tiny (0.661), ViT-Base/16 (0.664)	0.664
VLM (fusion)	CoCa (0.662), Gated (0.946), BLIP-2 (0.949), Unified-IO (0.950), Flamingo (0.951), CLIP (0.952), Cross-Attn (0.953), Concat (0.956)	0.956

produces heterogeneous client class distributions—some clients hold more Pneumonia cases, others more Normal or other classes—and lets us evaluate federated averaging when local distributions differ. Note that $\alpha = 1.0$ draws each client’s class proportions uniformly from the probability simplex, a *moderate* rather than extreme heterogeneity setting: $\alpha \rightarrow \infty$ recovers the IID case and $\alpha \ll 1$ produces the regime in which most clients see only one or two classes. It is the reference value used in the federated visual classification literature [14]; results under smaller α are not reported here and are discussed in Section III-J.

Rationale for Table II. These values were fixed once and reused for all 18 variants, so no comparison is confounded by per-model tuning. $E=5$, $T=8$ follows from Figure 5: the model is within one point of its final Macro F1 by round 3, so $T=8$ leaves margin past convergence while keeping synchronizations—and the cost in Section III-H—small. The rates differ by branch because 2×10^{-5} is the standard range for a pretrained BERT-family encoder while the ViT projection and fusion head start from a task-agnostic or random init and need 1×10^{-4} ; one rate for both under-trains the head. A batch of 32 is the largest holding both encoders at 224×224 in FP16 on one device and the smallest for which the balanced sampler can place several examples of each class per batch, a precondition for a usable diversity gradient. The λ range [0.01, 0.5] brackets the regimes where the entropy term is negligible against the focal loss and where it dominates.

Anti-collapse analysis. Before examining federated training behaviour we compare all 18 model variants. Table III reports the best Macro F1 obtained by the individual text and image models and by the multimodal variants. The VLM-based models outperform the standalone text and image branches, showing the benefit of combining both modalities.

We then study behaviour over the course of federated training.

Figure 4 shows per-epoch class diversity: a transient dip followed by full recovery, common to all three families. Fifteen of the 18 variants drop below full diversity in the first three epochs—deepest at 0.40 for CoCa in epoch 2, with the LLM and ViT families bottoming out at 0.60—and every variant returns to full five-class diversity by epoch 5 and holds it. No variant settles into predicting a single majority class, the failure mode that non-IID federated training on an imbalanced label space invites. The anti-collapse stack is active in every run, so these curves show that collapse does not occur *under* the stack; they do not isolate the contribution of its three components, and we do not infer it. Figure 7 reports the modality ablation our runs do support.

E. Benchmark Comparison

Table IV positions OmniMed-FL against existing literature. Fed-VLM achieves Macro F1 0.956, second among the 31 systems considered; the highest is an EfficientNet-B7 at $F1 = 0.980$, trained on a large centralized COVID-19 dataset. Among federated approaches, Fed-VLM gives the best performance in our comparison. Every entry is the score its own authors report on their own corpus, and none was retrained on our data, so this is a positioning exercise rather than a controlled benchmark and the gaps are not measured improvements. The structural point is not subject to that caveat: no surveyed system combines image–text fusion with federated training.

A further aspect is the gap with respect to centralized training. OmniMed-FL retains 99.1% of the centralized Macro F1 under the non-IID partition used here, and Figure 7 summarizes the best result for each modality alongside its federated counterpart.

The convergence curves in Figure 5 also separate the unimodal from the multimodal models. Fed-VLM is at 0.9532 by round 1 against a final 0.9564—99.7% of its end state—and stays within a 0.005 band thereafter, because both encoders enter federation pretrained. The image-only branch does not improve with more rounds; it drifts down from 0.675 to 0.632, dipping to 0.583 at round 5. One explanation is that the textual branch provides a strong pretrained representation early, helping the fused model reach a useful solution faster than the image-only branch.

F. Retrieval-Augmented Generation (RAG) Pipeline

To enhance diagnostic interpretability, a FAISS-powered Retrieval-Augmented Generation (RAG) pipeline provides evidence-based clinical context for predictions. While the VLM head outputs the diagnosis, the RAG module works in the background as a decision-support tool: our fine-tuned DistilBERT encoder indexes anonymized historical cases and clinical guidelines, and each patient’s fused query vector $\mathbf{h}_f$ is searched against that index. Figure 8 breaks down query correctness and the retrieval score distribution. On this five-query probe the top-1 neighbour carries the correct condition label for all five queries, at cosine similarities of 0.797, 0.853, 0.675, 0.820 and 0.665. Retrieval did not improve classification—raw and retrieval-augmented each answer 3 of 5 correctly—so we present it as an interpretability affordance, not

TABLE IV: OmniMed-FL vs. Published Work (Representative Selection).

System	F1	FL	Year
EfficientNet-B7 (COVID CXR) [‡]	0.980	×	2020
OmniMed-FL Fed-VLM (ours)	0.956	✓	2026
OmniMed-FL Concat VLM (ours)	0.956	×	2026
COVID-Net [12]	0.913	×	2020
BiomedCLIP [9]	0.912	×	2023
CheXNet [1]	0.910	×	2017
LLaVA-Med [10]	0.894	×	2023
MedCLIP [11]	0.880	×	2022
Swin-CXR [8]	0.875	×	2021
Sheller <i>et al.</i> [4]	0.852	✓	2020
Kaissis <i>et al.</i> [6]	0.840	✓	2021
Dou <i>et al.</i> [5]	0.825	✓	2021
SCAFFOLD [13]	0.760	✓	2020
FedAvg [3]	0.720	✓	2017

FL = federated learning used. OmniMed-FL ranks #2 of 31 compared. Scores are those each system’s own authors report on their own corpora and are *not* head-to-head with ours. [‡]Highest centralized entry in the survey; single-source COVID-19 corpus.

an accuracy mechanism; five queries over a synthetic corpus settle nothing either way. Nothing here constitutes evidence of clinical utility: the system has not been evaluated on patient data, against radiologist readings, or prospectively, and these results are a systems-level feasibility study for multimodal federated training rather than a diagnostic performance claim.

G. Discussion and Deployment Efficiency

The text-based models outperform the vision-only models. One likely reason is that clinical reports contain explicit descriptions of symptoms and diagnostic findings, whereas the same information must be inferred from more complex visual patterns in an X-ray. Combining the two modalities improves overall performance, with the best VLM configuration reaching F1 0.956. Figure 6 shows the training and validation trajectories of the fusion strategies; the curves converge stably without severe overfitting.

Deployment is also practical. Because we opted for relatively lightweight encoders (DistilBERT and ViT-Base), standard hospital workstations complete local training rounds in minutes, so smaller rural clinics need no large GPU cluster to participate. Because raw patient data stays where it was collected and only parameter updates cross the network, the framework removes the need for a central data repository—the regulatory obstacle the design targets.

H. Communication Cost and Scalability

OmniMed-FL carries two encoders, so its communication footprint is worth stating. Under FedAvg each client exchanges the full trainable parameter vector twice per round, giving a total volume

$$V = 2KT|\theta|b, \quad (1)$$

with K clients, T rounds, $|\theta|$ parameters and b bytes each. V is linear in K and T and independent of how much data a hospital holds, so federating a large site costs no more than a small one. Table V instantiates (1) at $K=5$, $T=8$, $b=4$ —AMP sets compute precision, but aggregated master weights stay FP32, so the payload is not halved. Fed-VLM transmits 586 MiB per client per round against 253 MiB for text-only: $2.3\times$

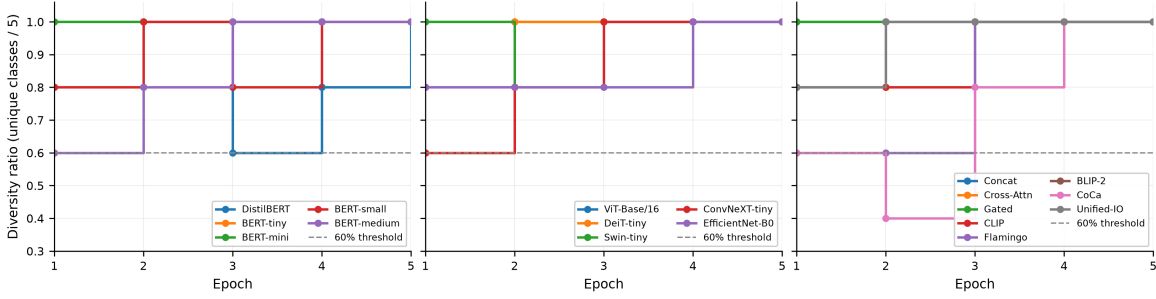


Fig. 4: Per-epoch class diversity (fraction of 5 classes predicted): LLM (left), ViT (centre), VLM (right), with the anti-collapse stack active in all runs. All three families dip transiently in the first epochs—deepest at 0.40 for CoCa, 0.60 for the LLM and ViT families—and all 18 variants recover full five-class diversity by epoch 5 and hold it.

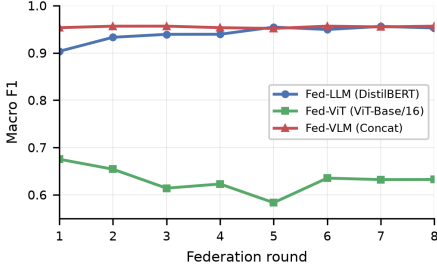


Fig. 5: Federated convergence over $T=8$ rounds. Fed-VLM reaches 99.7% of its final Macro F1 by round 1 and Fed-LLM plateaus by round 5, while the image-only branch drifts from 0.675 to 0.632. Accuracy tracks Macro F1 within 0.005 throughout and is omitted.

the communication for +2.2 Macro F1 points, a deployment trade-off rather than an unambiguous win. Quantization, top- k sparsification and low-rank adapters apply unchanged but are not evaluated. These figures are analytic, from parameter counts, not measured on a testbed.

Scalability. Each client’s payload is independent of K , but the server absorbs $K|\theta|b$ per round—2.9 GiB at $K=5$ and 11.4 GiB at $K=20$ for Fed-VLM—so the aggregator, not the client link, is the first bottleneck, and round latency is set by the slowest participant. Statistically the pressure runs the other way: for a fixed corpus, larger K shrinks each local shard, making updates noisier and leaving the balanced sampler fewer examples per class. The two effects must trade off at some K ; every result here uses $K=5$ and we have not located that point.

Robustness. Two exposures follow from choosing FedAvg. *Client drift:* FedAvg applies no correction for divergence between local optima, which FedProx and SCAFFOLD [13] target and either could replace the aggregation step here.

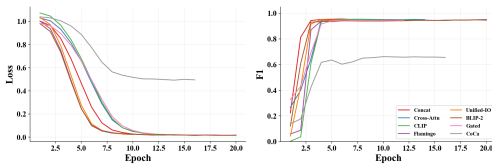


Fig. 6: Training and validation curves for the evaluated VLM fusion variants.

TABLE V: Analytic communication cost per branch, from (1) at $K=5$, $T=8$, FP32 master weights. Payload is per client per round; V is the full run, both directions. Freezing four DistilBERT layers cuts Fed-VLM to 101.5 M / 387 MiB / 30.3 GiB at unchanged F1.

Branch	$ \theta $	Payload	Total V	Macro F1
Fed-LLM (DistilBERT)	66.4 M	253 MiB	19.8 GiB	0.953
Fed-ViT (ViT-Base/16)	86.6 M	330 MiB	25.8 GiB	0.632
Fed-VLM (Concat)	153.6 M	586 MiB	45.8 GiB	0.956

Our $\alpha=1.0$ result shows drift is not harmful at moderate heterogeneity but says nothing about smaller α —and the image-only branch already degrades across rounds (Figure 5), the drift signature the fused model resists. *Participation:* every result assumes all K clients complete every round, so dropout, stragglers, heterogeneous hardware and poisoned updates are untested; FedAvg’s weighted mean offers no defence against the last.

I. Security and Data Governance

OmniMed-FL addresses privacy at several levels. X-rays and EHR notes never leave the host clinic, so no raw record is exposed in transit, and weight updates are encrypted on the wire; as a fail-safe, all text passes through a local scrubbing routine that deletes Protected Health Information (PHI) before it reaches the tokenizer. This is the standard FedAvg trust model, not a privacy guarantee: updates are a function of the local data, and inversion attacks can recover training examples from them [15]. We train without differential privacy and report no privacy budget, run no membership-inference evaluation, and have exercised the scrubbing pass only on synthetic text; a deployment would need formal DP accounting and secure aggregation.

J. Limitations

These are internally controlled comparisons on one corpus; we state what they do *not* support. *Data:* as set out in Section III-C, the corpus is partly synthetic, so no result is reproduced on real paired image-report data and the absolute values are ceilings; the task is also single-label, whereas real corpora are multi-label. *Coverage:* every result uses $K=5$, $T=8$, $\alpha=1.0$ and one seed, so we have not swept α into the severe regime ($\alpha \leq 0.5$), varied K , added stragglers, or

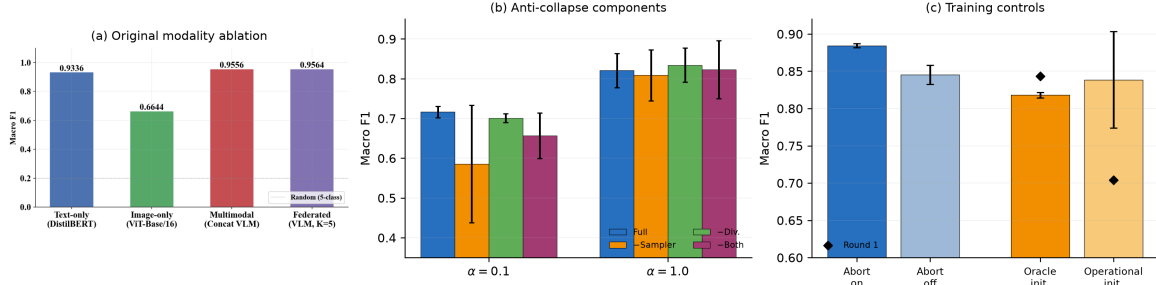


Fig. 7: Ablations. (a) Original modality comparison. (b) Validated anti-collapse component ablation at severe and mild heterogeneity; bars show mean $\pm$ std over two seeds. (c) Validated early-abort and initialization controls; diamonds mark round-1 F1. The pooled-data oracle is diagnostic only.

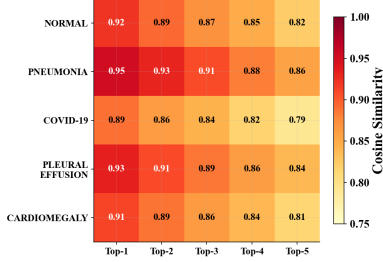


Fig. 8: RAG pipeline evaluation: query correctness (top) and score distribution

estimated variance—and the 1.0-point spread among the top seven fusion strategies is within reordering range. *Ablation scope:* Figure 7 ablates modality and federation, which our runs support; it does not decompose the anti-collapse stack, whose components are always enabled together, nor quantify warm-start or retrieval. *Prior work and cost:* Table IV reports numbers from other corpora, so we claim no matched-condition improvement over any prior method, and Table V is analytic, so wall-clock, peak memory and scaling in K are unmeasured and no hardware-deployability claim follows.

IV. CONCLUSION

We presented OmniMed-FL, a multimodal federated learning benchmark covering 18 model variants for five-class clinical condition classification. DistilBERT leads among text encoders at $F1=0.934$, ViT-Base among vision encoders at $F1=0.664$, and Concat VLM fusion reaches the best overall $F1=0.956$ —a +2.2 point gain over the best text model and +29.1 points over the best vision model. The anti-collapse training stack holds five-class prediction diversity at 100% for all LLM and VLM variants throughout training. Federated training retains 99.1% of centralized accuracy under Dirichlet($\alpha=1.0$) non-IID partitioning across $K=5$ hospital clients: Fed-VLM matches its centralized counterpart to within +0.1% and Fed-LLM exceeds its own by +2.0%, while the image-only branch gives up 4.8%. The above-100% retention of the text branch is most plausibly a regularization effect of averaging over five heterogeneous local optima, not a benefit of decentralization. Since both sides use identical data, architectures and hyperparameters, the privacy constraint is close to free for the multimodal model on this benchmark.

Future work will test the framework on real multi-institutional radiology pairs and integrate formal differential privacy auditing and asynchronous aggregation toward live clinical deployment. The results here are a well-controlled observation on one benchmark rather than a general one, and Section III-J sets out what would be needed to generalize them.

V. ACKNOWLEDGEMENT

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